

Source-Grounded Study Notes Pipeline

Author: Masoom Sakina

Date: August 19, 2026

Track: AI Fluency / Machine Learning

1. Workflow Diagram

[Raw PDF Paper] → (NotebookLM) *Action: Source-Grounded Extraction*
→ [Extracted Methodology] → (Claude Project) *Action: Critical Synthesis & Leakage Check*
→ [Critical Analysis] → (Claude Project) *Action: Standardized Formatting*
→ [Final Markdown Study Note]

2. Prompts & Configuration

Step 1: Extractor (NotebookLM)

Act as a senior quantitative researcher. Extract the core machine learning methodology from this paper. Specifically identify: 1) The ensemble models utilized. 2) The exact feature engineering or temporal data handling techniques used. 3) How SHAP or other explainability metrics were applied. Quote the specific metrics directly from the text. Do not hallucinate external information.

Step 2: Synthesizer (Claude Project)

I am building an Intelligent Framework For Interpretable Time Series. Here is the methodology extracted from a recent financial machine learning paper: [PASTE NOTEBOOKLM TEXT HERE]. Critique this approach. What are the potential target leakage risks in their temporal handling? Are there weaknesses in their SHAP implementation that I should avoid in my own pipeline?

Step 3: Formatter (Claude Project)

Convert your critique into a strict Markdown table with the following headers: Paper Focus, Models Used, Explainability Strategy, Major Weakness / Leakage Risk, and Application to My Capstone.

3. Pipeline Runs (5 Real Inputs)

Run 1: AFRN-HyperFlow Framework

Paper Focus	Models Used	Explainability Strategy	Major Weakness / Leakage Risk	Application to My Capstone
Adaptive ensemble framework for financial time-series.	ESN, TCN, MDN, Hypernetwork, DSSM.	Systematic SHAP value analysis for directional impact + temporal attention.	Leakage Control: Strict 5-day gap. Weakness: Massive computational overhead making real-time SHAP too expensive.	Adopt 5-day gap for out-of-domain validation. Use LightGBM/RF for faster SHAP.

Run 2: Sovereign Downgrade Prediction

Paper Focus	Models Used	Explainability Strategy	Major Weakness / Leakage Risk	Application to My Capstone
Predicting sovereign credit downgrades using a glass box approach.	XGBoost, Logistic Regression.	SHAP global feature importance and local case-study risk audits.	Leakage Control: Strict t-1 lag and temporal split. Weakness: Low-frequency annual data makes it a coincident indicator.	Implement 3-Year Rolling Mean and Std Dev for trend/volatility capture in tabular contracts.

Run 3: Online Ensemble for Sector Rotation

Paper Focus	Models Used	Explainability Strategy	Major Weakness / Leakage Risk	Application to My Capstone
Sector rotation and portfolio allocation via online learning.	Multiplicative Weights Update (MWU) over technical experts.	Innate transparency via weight allocation observation (no SHAP).	Leakage Control: Updates at t use loss from t-1 (no look-ahead). Weakness: Cannot explain *why* base models act; slow adaptation.	Adapt sequential updating architecture to prevent target leakage in the ensemble layer.

Run 4: SHAP-Based Comparative Analysis

Paper Focus	Models Used	Explainability Strategy	Major Weakness / Leakage Risk	Application to My Capstone
Evaluating model interpretations of financial features.	Random Forest, Gradient Boosting, SVM, Neural Networks.	SHAP for global rankings and local explanations across models.	Weakness: Multicollinearity makes SHAP unstable. Leakage Risk: Standard k-fold	Implement strict collinearity management (VIF checks) before modeling for stable SHAP.

			causes SHAP to explain future data.	
--	--	--	-------------------------------------	--

Run 5: Ensemble Approach to Stock Price Prediction

Paper Focus	Models Used	Explainability Strategy	Major Weakness / Leakage Risk	Application to My Capstone
Hybrid ensemble optimized with custom financial loss functions.	LSTM, GRU, Linear Regression, LightGBM.	None explicitly defined (opaque black boxes).	Leakage Risk: Global feature scaling introduces severe look-ahead bias. Weakness: Lacks adaptive weighting.	Incorporate Sharpe Ratio/Correlation Loss but strictly enforce rolling-window scaling.

4. Time Accounting & Failure Points

Time Accounting:

- **Setup Cost:** ~30 minutes (Designing architecture, prompts, and selecting papers).
- **Pipeline Execution:** ~15 minutes (~3 mins per paper x 5 papers).
- **Total Time Spent:** 45 minutes.
- **Time Saved:** A manual review and synthesis of these 5 dense quantitative finance papers would take 4 to 5 hours. **Total saved:** ~4 hours.

Known Failure Points & Required Human Review:

- **Math Hallucinations:** The AI pipeline struggles to correctly parse and reproduce complex mathematical loss functions and mathematical proofs from PDFs. Human review is required to verify equations.
- **Financial Nuance Over-Simplification:** The critique sometimes glosses over micro-structure market behaviors. A human must review the generated tables to ensure the specific trading context is accurate.